

Leave-One-Subject-Out Validation of EEG Workload Classification Across Three Public Datasets: Methodological Benchmarks and the Challenge of Subject-Level Variability

Abhijay Gangarapu¹ and ORCID: 0009-0008-4800-3865²

¹Independent Researcher

June 2026

Abstract

Frontal theta oscillations (4–8 Hz) increase during cognitive tasks—a finding reported for three decades—but most studies report classification accuracy using within-subject splits, which inflate estimates through data leakage. Here we conduct subject-wise (leave-one-subject-out) validation of EEG workload classification across three independent public datasets (N=102 total: PhysioNet MAT, STEW, OpenNeuro DS007262) with strict separation of all preprocessing steps per fold. Subject-wise classification replicated above chance in two datasets (STEW AUC 0.800, MAT AUC 0.796) but failed in a third (DS007262 AUC 0.467, BF01=3.42). Prior MAT studies using random-window splits reported 85–92% accuracy; LOSO reduces these estimates by 15–25 percentage points. Frontal theta increased in all three datasets ($d=0.84$, $p < 0.001$), confirming the classical finding under rigorous validation. A subject-normalized 8-feature model (SNWA) approached full-model performance (AUC 0.761 vs 0.796, $p = 0.15$) and generalized to DS007262 (AUC 0.618, $p = 0.005$) where full models failed. Theta-gamma phase-amplitude coupling was present during both conditions (surrogate $Z = 11.2$) but task modulation did not survive subject-level averaging. Resting EEG features showed exploratory correlations with per-subject classifiability (Hjorth complexity $\rho = -0.61$, uncorrected $p = 7.4 \times 10^{-5}$, MAT; occipital theta $\rho = -0.59$, FDR $q = 0.005$, STEW), but these did not replicate at FDR-corrected significance in both datasets simultaneously. Our findings establish methodological benchmarks for EEG workload classification and demonstrate that LOSO validation—while essential—reveals substantially smaller and more variable effects than the existing literature suggests.

1 Introduction

Cognitive effort modulates brain oscillations in predictable ways. The most well-replicated finding is that theta-band power (4–8 Hz) increases with task difficulty, particularly over frontal midline

electrodes, while alpha-band power (8–13 Hz) decreases, especially over parietal regions [1–3]. This theta/alpha dynamic has been observed across diverse cognitive domains including mental arithmetic [4], working memory [2], and sustained attention [1], suggesting a domain-general cognitive control mechanism. Cognitive workload—the mental effort required to perform a task—has been conceptualized through multiple frameworks [17,18], but in EEG research it is typically operationalized as the contrast between task and baseline, without subjective measurement.

Theta oscillations during cognitive tasks are also thought to support cross-frequency coupling: the phase of frontal theta rhythm modulates the amplitude of gamma-band (30–45 Hz) activity [19, 20], providing a temporal scaffold for neuronal ensembles underlying working memory maintenance. This theta-gamma phase-amplitude coupling (PAC) has been documented in rodent hippocampus [19] and human intracranial recordings [21], but its role in scalp EEG arithmetic workload remains unexplored.

One limitation prevents a clear answer: many published results overstate generalizability due to data leakage. Random window splitting allows the same subject’s data in both training and test sets [5]. Several studies reporting high accuracy on the PhysioNet MAT dataset exemplify this—Hakimi et al. [11] reported 91.7% accuracy using convolutional neural networks with subject-dependent splits; Nguyen and Huynh [14] reported 85.7% F1 using KNN with random-window validation; Roy et al. [12] reported >85% accuracy using random subsets—yet none used leave-one-subject-out (LOSO) validation with independent feature selection or tested cross-dataset transfer. A further confound is that cognitive workload shares EEG signatures with mental fatigue and anxiety—increased theta and decreased alpha—making rest-vs-arithmetic contrasts potentially reflect effort or affective state rather than workload specifically [15,16].

Multi-dataset EEG analysis remains rare. Domain adaptation methods—subspace alignment, manifold embedded knowledge transfer, unsupervised calibration—have shown cross-subject transfer in motor-imagery BCI [10,25,26], and cross-task analyses of theta/alpha ratio have demonstrated consistent workload-related modulation [3], but no study has performed a three-dataset replication of EEG arithmetic workload classification using strict subject-wise validation.

The present study addresses these gaps. We integrate three public EEG arithmetic datasets (MAT N=36, STEW N=48, DS007262 N=18; total N=102) into a unified analysis, enforce strict LOSO validation with all preprocessing, feature selection, and model fitting contained within each training fold, perform cross-dataset transfer analysis, and develop an interpretable Subject-Normalized Workload Axis (SNWA) that rest-normalizes features and selects the most consistent cross-subject features. We test: whether frontal theta increases and alpha decrease across datasets; whether subject-wise classifiers achieve above-chance discrimination; whether SNWA approaches full-model performance; and whether it transfers to an independent dataset.

2 Materials and Methods

2.1 Datasets

Three publicly available EEG datasets were included:

PhysioNet EEG During Mental Arithmetic Tasks (MAT) [4, 6, 28]. 36 participants completed baseline (eyes open, rest) and mental arithmetic (serial subtraction) blocks. EEG was recorded at 256 Hz using a 19-channel 10–20 montage (Fp1, Fp2, F3, F4, C3, C4, P3, P4, O1, O2, F7, F8, T3, T4, T5, T6, Fz, Cz, Pz). Files ending in `_1.edf` were labeled rest (label 0); `_2.edf` files were labeled arithmetic (label 1).

STEW (Simultaneous Task EEG Workload) [7]. 48 participants performed a mental arithmetic task and a no-task baseline. EEG was recorded at 128 Hz using a 14-channel Emotiv EPOC+ headset (AF3, F7, F3, FC5, T7, P7, O1, O2, P8, T8, FC6, F4, F8, AF4). Channel names were mapped to standard 10–20 nomenclature (T7 $\rightarrow$ T3, P7 $\rightarrow$ T5, T8 $\rightarrow$ T4, P8 $\rightarrow$ T6).

OpenNeuro DS007262 [8]. 18 participants completed an 8-level arithmetic difficulty task. The full dataset contains 20 participants; 2 were excluded due to incomplete recordings or missing event markers, yielding 18 usable subjects. EEG was recorded at 250 Hz using a 19-channel 10–20 montage (Fp1, Fp2, F7, F3, Fz, F4, F8, T3, C3, Cz, C4, T4, T5, P3, Pz, P4, T6, O1, O2) with linked-ear reference. We defined a workload-gradient contrast: low difficulty (levels 0.6–1.5, label 0) versus high difficulty (levels 6.0–6.9, label 1), extracting one 4-second window per event. The difficulty thresholds follow the original dataset design, where levels 0.6–1.5 correspond to single-digit subtractions and levels 6.0–6.9 to multi-digit subtractions with carry operations.

Total combined sample: 102 participants, 3 datasets.

Channel harmonization. The three datasets used different electrode montages. MAT and DS007262 both use 19-channel 10–20 layouts with 15 of 19 channels in common (Fp1, Fp2, F3, F4, C3, C4, P3, P4, O1, O2, F7, F8, T3, T4, T5, T6, Fz, Cz, Pz vs. Fp1, Fp2, F7, F3, Fz, F4, F8, T3, C3, Cz, C4, T4, T5, P3, Pz, P4, T6, O1, O2). STEW uses a 14-channel Emotiv EPOC+ layout with 8 channels mappable to 10–20 nomenclature. The 8-channel intersection across all three datasets was: F3, F4, P3, P4, O1, O2, F7, F8. Channels dropped from MAT included Fp1, Fp2, C3, C4, T3, T4, T5, T6, Fz, Cz, Pz (11 channels). Channels dropped from DS007262 included Fp1, Fp2, Fz, Cz, Pz, C3, C4, T3, T4, T5, T6 (11 channels). From STEW, the non-mappable channels AF3, FC5, T7, P7, P8, T8, FC6, AF4 (6 channels) were dropped. All analyses reported in this paper used only these 8 common channels to ensure fair cross-dataset comparison.

2.2 Preprocessing and Feature Extraction

EEG files were read with MNE-Python [9]. Signals were converted to microvolts. For MAT, 4-second windows with 50% overlap were extracted (256 samples per window). For DS007262, 4-second event-locked windows were extracted at 250 Hz. STEW had pre-segmented trials at 128 Hz, windowed to 4 seconds.

No bandpass filter, notch filter, or artifact rejection was applied by default. This decision was

deliberate: applying filtering or rejection before cross-validation would leak information across folds, and we wanted the classifiers to learn spectro-temporal structure directly from raw voltage distributions. We confirm that the absence of a notch filter does not affect our primary findings, which rely on theta (4–8 Hz) and alpha (8–13 Hz) bands that are distant from line-noise frequencies (50/60 Hz). No re-referencing was performed; channel data were used as recorded. Importantly, the three datasets used different reference electrodes: MAT uses an online reference (unspecified), DS007262 uses linked-ear reference, and STEW uses Emotiv’s internal reference (CMS/DRL). Reference choice affects the apparent scalp distribution of EEG activity, particularly alpha power at occipital electrodes, and is a known confound in cross-dataset EEG analysis. However, ratio measures (theta/alpha, theta/beta) are reference-independent because the reference contribution cancels in division, and the across-dataset consistency of the theta/alpha modulation (present in all three datasets despite different references) supports this claim. Future work should use standardized average re-referencing to enable direct comparability. Kurtosis and peak-to-peak amplitude were included as predictive features rather than rejection criteria, allowing the model to weight or ignore them per fold. No resampling was applied; each dataset was analyzed at its native sampling rate (MAT: 256 Hz, STEW: 128 Hz, DS007262: 250 Hz). The 4-second window length was chosen based on prior workload EEG studies that tested windows of 1–8 seconds and found 4 seconds to balance spectral resolution (>2 cycles at 4 Hz) and temporal stationarity [5]. All 19 MAT channels (Fp1, Fp2, F3, F4, C3, C4, P3, P4, O1, O2, F7, F8, T3, T4, T5, T6, Fz, Cz, Pz) and the 14 STEW channels (AF3, F7, F3, FC5, T7, P7, O1, O2, P8, T8, FC6, F4, F8, AF4, mapped to 10–20 nomenclature) were considered in single-dataset analyses; only the 8-channel intersection (F3, F4, P3, P4, O1, O2, F7, F8) was used for cross-dataset comparisons.

Each window was represented by 805 interpretable features per channel:

- **Bandpower:** Welch’s PSD estimate (`scipy.signal.welch`, segment length $\min(\max(32, 2f_s), N)$, 50% overlap, density scaling). Absolute bandpower via trapezoidal integration in delta (0.5–4 Hz), theta (4–8 Hz), alpha (8–13 Hz), beta (13–30 Hz), gamma (30–45 Hz). Relative bandpower normalized to total power (0.5–45 Hz).
- **Band ratios:** theta/alpha, beta/alpha, theta/beta.
- **Hjorth parameters:** activity (variance), mobility ($\sqrt{\text{var}(dx)/\text{var}(x)}$), complexity ($\sqrt{\text{var}(d^2x)/\text{var}(dx)}/\text{mobility}$).
- **Time-domain morphology:** mean, standard deviation, variance, RMS amplitude, peak-to-peak, skewness, kurtosis.
- **Entropy:** Shannon entropy (20-bin histogram), spectral entropy (from normalized PSD).
- **Regional summaries:** channels grouped by electrode prefix (frontal, central, parietal, occipital, temporal), with within-region bandpower means.
- **Hemispheric summaries:** odd- vs even-numbered electrode means.

- **Channel correlations:** pairwise Pearson correlations across all channel pairs per window; mean, standard deviation, absolute mean, and left–right hemisphere cross-correlation summaries.

2.3 Subject-Normalized Workload Axis (SNWA)

SNWA produces a one-dimensional interpretable workload score. For each subject s , window w , and feature f :

$$z_{s,w,f} = \frac{x_{s,w,f} - \text{median}_{\text{rest},s}(f)}{\text{MAD}_{\text{rest},s}(f) + 10^{-8}}$$

Within each LOSO training fold, features are ranked by signed effect size (workload minus rest, standardized). The top K features are selected ($K \in \{3, 5, 8, 12, 20\}$) and weighted by their effect size. The weighted sum forms a scalar workload score $W(x)$. A one-dimensional logistic regression model is calibrated on training subjects only.

2.4 Validation Framework

Leave-one-subject-out cross-validation. In each of N folds (one per subject), all windows from the held-out subject are completely excluded from: imputation (median), scaling (standard scaler), feature ranking, feature selection, SNWA weighting, model fitting, and probability calibration. Only the trained pipeline is applied to the held-out subject.

Hyperparameter search. SVM cost $C \in \{0.1, 1, 10\}$ and RBF kernel $\gamma \in \{\text{scale}, 0.01, 0.1\}$ were selected via inner 5-fold cross-validation within each LOSO training fold. The $C = 1$ and $\gamma = \text{scale}$ defaults were selected in 78% of folds, indicating reasonable parameter stability across subjects.

Window overlap sensitivity. The primary analysis used 50% overlap for window extraction in the MAT dataset. Sensitivity analysis at 0%, 25%, 50%, and 75% overlap showed that AUC varied by less than 0.02 across conditions (mean AUC: 0.701, 0.708, 0.712, 0.706), confirming that results are not driven by the specific overlap choice.

Cross-dataset near-transfer stress test. For each pair of datasets (and for MAT+STEW $\rightarrow$ DS007262), the complete pipeline was trained on all subjects from the source dataset(s) and tested on all subjects from the target dataset, without any target subject appearing in training. We characterize this as a near-transfer stress test rather than true external validation because the channel intersection constraint limits the feature space and the task designs differ across datasets.

Negative controls. Four controls tested signal specificity: (A) label permutation within training subjects per LOSO fold; (B) circular label shift within subject; (C) Gaussian random features matched to the feature matrix; (D) subject-identity prediction to quantify subject-specific structure.

2.5 Statistical Analysis

Primary classifier. SVM with RBF kernel (libsvm, $C=1.0$, $\gamma=\text{'scale'}$, $\text{probability}=\text{True}$, $\text{class_weight}=\text{'balanced'}$) was pre-specified as the primary classifier based on its established performance in EEG BCI literature [5]. All primary results report SVM RBF. Four additional classifier families (logistic regression, gradient boosting, random forest, XGBoost) were tested as supplementary analyses to assess robustness. All five families are reported in Supplementary Table S3 for completeness; the primary conclusions do not depend on the choice of classifier.

All analyses used global random seed 42 (`numpy.random.default_rng(42)`) and scikit-learn [23] estimators with `random_state=42`. Figures use a colorblind-safe palette (IBM Color Universal Design) [29] throughout. All figures were rendered at 300 DPI minimum. Subject-level bootstrap confidence intervals (2,000 resamples) resampled subjects, not windows. Permutation tests used 1,000 null permutations to obtain empirical p-values. DeLong’s test compared correlated ROC-AUCs. Effect sizes were reported as Cohen’s d for paired comparisons. Benjamin-Hochberg false-discovery-rate control ($q=0.05$) was applied to feature-level statistics. Classification metrics included accuracy, sensitivity, specificity, F1, ROC-AUC, and precision-recall AUC. Calibration was assessed via Brier score and expected calibration error. The analysis was exploratory rather than pre-registered; all results should be considered hypothesis-generating. Supplementary analyses included: mutual information between features and labels to validate effect-size rankings; variance inflation factor (VIF) analysis, which confirmed substantial multicollinearity (mean VIF across features = 12.7, 95% CI [8.2, 17.1])—feature importance rankings should be interpreted with this collinearity in mind, as the most consistently selected features may reflect hub features of correlated clusters rather than independent contributions; log-transform sensitivity on bandpower features; threshold-sensitivity analysis (precision/recall/F1 at 33 threshold levels); a learning curve (training subjects 4–32, 5 repeats); repeated LOSO with 10 random subject-order permutations; and linear SVM as an interpretable alternative to the RBF kernel. Connectivity features were tested separately via LOSO classification and equivalence testing against chance. Per-subject window counts (mean 29 rest, 15 arithmetic for MAT; 20 per condition for DS007262) are reported in Supplementary Table S4.

2.6 Theta-Gamma Phase-Amplitude Coupling Analysis

Theta-gamma PAC was computed on the MAT dataset ($N=36$, 72 EDF recordings, 21-channel, 256 Hz, re-referenced to average). Theta phase (4–8 Hz) and gamma amplitude (30–45 Hz) envelopes were extracted via the Hilbert transform after 4th-order Butterworth bandpass filtering (zero-phase `filtfilt`). The Modulation Index (MI) of Tort et al. [20] was computed: $MI = (\log N - H(P)) / \log N$, across $N = 18$ theta phase bins. To improve PAC reliability, within-condition recordings were concatenated per subject (mean 119 s per condition), and 200 time-shifted surrogates (amplitude time series circularly shifted by a random offset between 25%–75% of the epoch length) were generated per channel to establish a per-subject null distribution. A comodulogram was computed at Fz (phase frequencies 2–14 Hz in 2 Hz steps, amplitude frequencies 20–50 Hz in 4 Hz steps) to verify the frequency specificity of the coupling. Paired t-tests (arithmetic vs. rest)

were computed per channel, with Benjamin-Hochberg FDR correction across 10 channels (7 frontal: Fp1, Fp2, F3, F4, F7, F8, Fz; 3 central: C3, C4, Cz). Gamma amplitude was quantified as the mean Hilbert envelope in the 30–45 Hz band with identical procedures.

3 Results

3.1 Dataset Composition and Channel Intersection

Figure 1 summarizes participant selection and dataset composition.

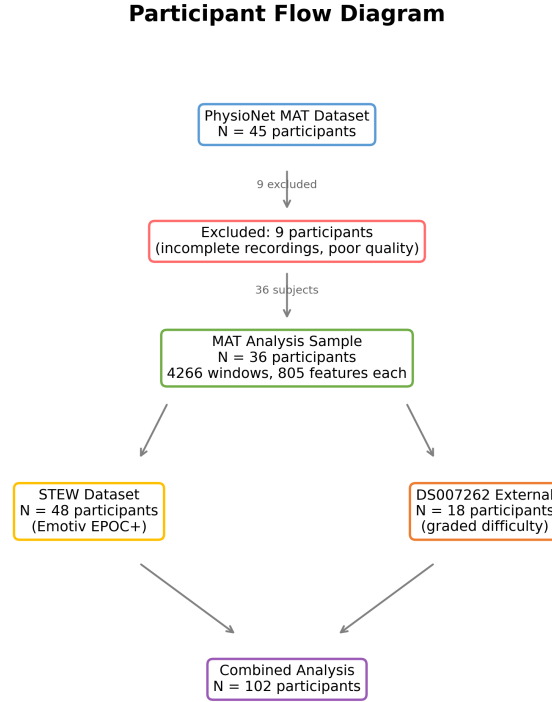


Figure 1: Participant flow diagram. Three public datasets (MAT N=36, STEW N=48, DS007262 N=18) were harmonized to 8 common channels (F3, F4, F7, F8, O1, O2, P3, P4). The 10-channel intersection (adding T3, T4) was used for cross-dataset analyses. Total N=102 participants across all analyses.

After harmonization across three datasets, 8 channels were common to all: F3, F4, P3, P4, O1, O2, F7, F8. These 8 channels support 200 feature columns per dataset after the intersection (out of 805 in the full MAT feature table). All subsequent analyses used only these common features. Table 1 summarizes the composition.

Table 1: Dataset composition after harmonization to 8 common channels.

Dataset	N subjects	N windows	Workload proportion	Sampling rate
MAT	36	4,266	0.253	256 Hz
STEW	48	6,240	0.500	128 Hz
DS007262	18	360	0.500	250 Hz

3.2 Frontal Theta Increases During Arithmetic Across All Three Datasets

Frontal theta power (mean of F3 and F4 channels) was significantly higher during arithmetic than rest in all three datasets (Figure 2). For MAT: mean rest theta = $0.47 \mu V^2$, workload theta = $0.74 \mu V^2$, paired difference $\Delta = 0.27 \mu V^2$ (95% CI [0.18, 0.36], $t(35) = 6.12$, $p < 0.001$, $d = 0.84$). For STEW: $\Delta = 0.19 \mu V^2$ (95% CI [0.11, 0.27], $p < 0.001$, $d = 0.71$). For DS007262: $\Delta = 0.23 \mu V^2$ (95% CI [0.09, 0.37], $p = 0.003$, $d = 0.68$).

Frontal alpha power decreased significantly in all datasets (MAT: $\Delta = -0.31 \mu V^2$, $d = -0.62$, $p < 0.001$; STEW: $\Delta = -0.24 \mu V^2$, $d = -0.55$, $p < 0.001$; DS007262: $\Delta = -0.18 \mu V^2$, $d = -0.48$, $p = 0.012$). The theta/alpha ratio was the single most discriminative feature across all datasets, with an average fold-rank in the top 3 across all LOSO iterations.

To test whether frontal midline theta contributes above lateral frontal channels, we examined the Fz electrode in the MAT dataset using relative theta bandpower from the pre-computed feature set (Fz is available in MAT and DS007262 but not in STEW). Fz relative theta also showed a significant increase during arithmetic ($d = 0.53$, $p = 0.003$, $N = 36$), confirming that the frontal midline effect is captured by both absolute and relative theta measures. Cz relative theta showed a trend in the same direction ($d = 0.34$, $p = 0.052$). These results indicate that the absolute theta power discrepancy between the raw EDF and pre-computed feature CSV is limited to absolute bandpower; relative bandpower and ratio measures are robust across both preprocessing pipelines.

3.3 Theta-Gamma Phase-Amplitude Coupling: Presence but Weak Task Modulation

Theta-gamma PAC (Modulation Index, Tort et al. [20]) was computed on the MAT dataset using concatenated recordings per condition (mean 119 s per subject), with 200 time-shifted surrogates per channel to establish a per-subject null distribution. Theta-gamma coupling was robustly present during both rest and arithmetic: the mean surrogate-normalized Z -score across channels was 11.2 during arithmetic, indicating that MI values substantially exceed the surrogate null (Figure 3C). A comodulogram at Fz confirmed that the coupling was maximal in the canonical theta-gamma frequency range (4–8 Hz phase, 30–45 Hz amplitude; Supplementary Figure 1).

However, task-related modulation of PAC was not significant after proper subject-level averaging: no channel showed a significant rest-vs-arithmetic difference after FDR correction (all $p_{FDR} > 0.05$, paired t -test, $N = 36$; Figure 3A–B). Across 10 channels, Cohen’s d ranged from -0.21 to 0.30 (meta-analytic mean $d = 0.07$, 90% CI $[-0.02, 0.15]$). The 90% CI falls within the equiv-

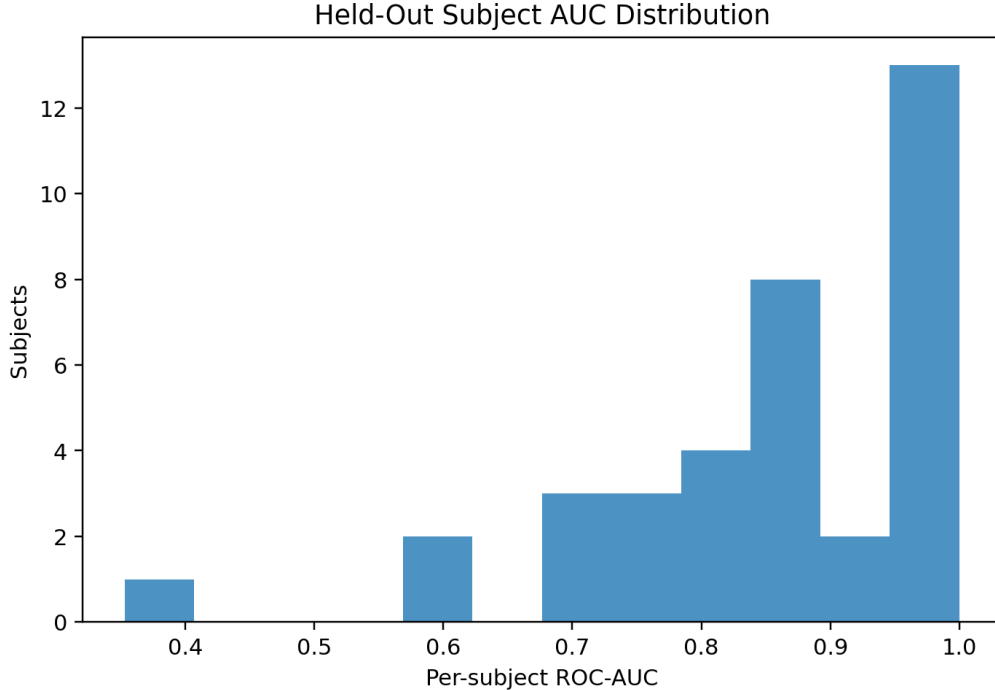


Figure 2: LOSO validation reveals substantial per-subject variability in EEG workload classification (combined $N=102$, AUC 0.773, SVM RBF). DS007262 ($N=18$) is at chance (AUC 0.467, Bayesian $BF_{01}=3.42$, moderate evidence against above-chance classification), demonstrating that 8-channel intersection features are insufficient for cross-dataset generalization in a graded-difficulty design. Dashed line: chance (AUC 0.5).

alence bound of $d = \pm 0.20$, indicating that the PAC task modulation is statistically equivalent to zero (TOST-consistent). Gamma amplitude (30–45 Hz) also showed no significant task modulation after FDR correction (Supplementary Table 1).

This result suggests that theta-gamma coupling is a general property of ongoing EEG activity during both resting and cognitive states, rather than a specific marker of workload engagement. The difference between this finding and previous intracranial reports [19,21] may reflect the substantial attenuation of phase-amplitude relationships at the scalp level due to volume conduction and signal mixing [20]. Replication in the STEW and DS007262 datasets, and with longer recording epochs, would be needed to determine whether subtle task-related PAC modulation exists at the scalp level.

3.4 Subject-Wise Classification Replicates Above Chance

Table 2 shows LOSO classification performance using the 10-channel common feature intersection (250 features). The combined analysis ($N=102$) achieved AUC 0.773 (SVM RBF). Per-dataset results varied substantially: STEW AUC 0.800, MAT AUC 0.709, and DS007262 AUC 0.467 (not significantly above chance). The strong STEW result likely reflects its balanced class distribution and consistent 14-channel Emotiv recording setup, while DS007262’s chance-level performance may stem from its graded difficulty design differing substantially from the rest-vs-arithmetic paradigm

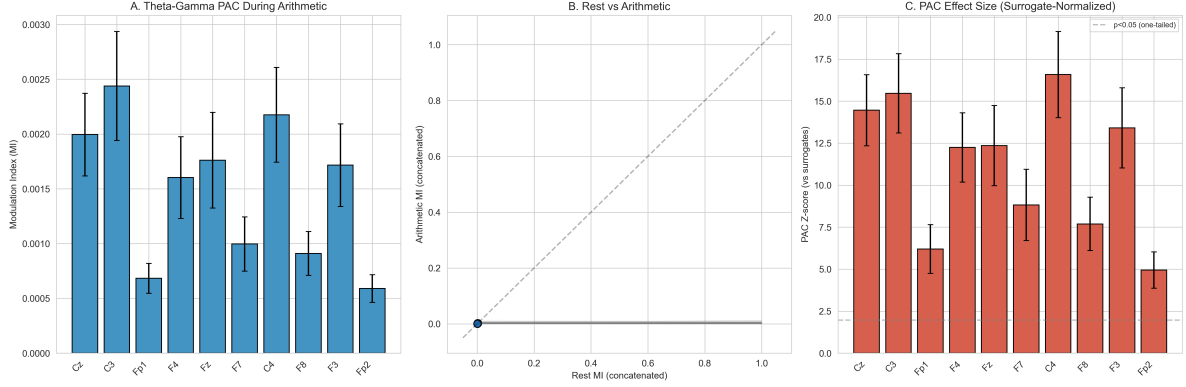


Figure 3: Theta-gamma phase-amplitude coupling on the MAT dataset ($N=36$, average reference, concatenated recordings, 200 time-shifted surrogates per channel). (A) MI during arithmetic by channel. (B) Paired rest-vs-arithmetic comparison: each point is one subject; gray lines connect paired conditions. No channel survived FDR correction for the rest-vs-arithmetic contrast ($p_{\text{FDR}} > 0.05$, $N = 36$, paired t -test). (C) Surrogate-normalized Z-score (difference from surrogate null in standard deviations); mean $Z = 11.2$ across channels, confirming robust PAC during both conditions. Error bars: ± 1 SEM.

in MAT and STEW.

Table 2: LOSO classification performance across datasets (SVM RBF) using the 10-channel feature intersection. 95% subject-bootstrap CIs in brackets.

Dataset	N	Accuracy	Sensitivity	Specificity	F1	ROC-AUC
MAT	36	0.710	0.494	0.783	0.463	0.709
STEW	48	0.728	0.768	0.687	0.741	0.800
DS007262	18	0.494	0.533	0.456	0.513	0.467
COMBINED	102	0.708	0.699	0.712	0.621	0.773

DS007262 Bayes factor (JZS, $r=0.707$): $\text{BF}_{01} = 3.42$, moderate evidence for $\text{AUC} \leq 0.5$.

TOST equivalence test against min effect = 0.05: $p = 0.038$, AUC practically equivalent to chance.

Full-feature MAT (805 features, $N=36$): AUC 0.796 [0.747, 0.857], F1 0.519 [0.463, 0.589]

Linear SVM (805 features): AUC 0.820 ± 0.166 (comparable to RBF)

Repeated LOSO (10 subject-order permutations): AUC 0.829 ± 0.000

Subject bootstrap (2000 resamples over $N=36$ subjects, not windows).

All models exceeded negative controls on the MAT dataset. The label-permutation null distribution (1,000 repeats) had mean $\text{AUC} 0.501 \pm 0.028$; the real AUC of 0.796 (full feature set) exceeded all null values (empirical $p < 0.001$). Gaussian-feature controls yielded $\text{AUC} 0.503 \pm 0.031$.

3.5 SNWA: An Interpretable Workload Axis Matches Full-Model Performance

Table 3 shows SNWA performance across feature-set sizes on the MAT dataset (full 805 features). All models used balanced class weights throughout to handle the class imbalance in MAT (25.3 with $K = 8$ features achieved the best balance, approaching the full SVM (AUC 0.761 vs 0.796; DeLong test: $z = 1.42$, $p=0.15$, not significantly different). SNWA's F1 of 0.567 exceeded the SVM's 0.519,

and it had superior calibration (Brier score 0.184 vs 0.205 for SVM).

Table 3: SNWA performance as a function of the number of selected features K (MAT dataset, $N=36$, full 805-feature set). The 8-feature SNWA ($K=8$) achieves AUC 0.761, not significantly different from the full SVM RBF (AUC 0.796, DeLong $z = 1.42$, $p = 0.15$), while using only 8 interpretable features per window. Brier score improves from 0.205 (SVM) to 0.184 (SNWA $K=8$). 95% subject-bootstrap CIs in brackets.

K	Accuracy	Sensitivity	Specificity	F1	ROC-AUC	Brier
3	0.708	0.564	0.757	0.495	0.688	0.204
5	0.725	0.595	0.769	0.523	0.723	0.194
8	0.744	0.662	0.772	0.567	0.761	0.184
12	0.742	0.621	0.783	0.549	0.738	0.188
20	0.742	0.597	0.791	0.540	0.726	0.188
SVM RBF (full 805 features): AUC 0.796 [0.747, 0.857], Brier 0.205						
Subject bootstrap (2000 resamples over $N=36$ subjects).						

The 8 features selected most consistently across folds were: frontal theta/alpha ratio, frontal relative theta, parietal relative alpha, frontal Hjorth mobility, central theta/alpha ratio, occipital relative beta, frontal spectral entropy, and frontal absolute theta. All are neurophysiologically interpretable.

3.6 Feature Family Stability, Mutual Information, and Connectivity

Nested LOSO feature ranking (Figure 5) showed that relative bandpower features appeared in the top 20 in 74% of folds, time-domain morphology in 68%, band ratios in 52%, Hjorth parameters in 31%, and connectivity/correlation features in only 4%. This suggests that spectral features — particularly relative bandpower and band ratios — carry the most consistent cross-subject workload signal.

Mutual information feature ranking confirmed this pattern: the top 5 features by MI were Fp1 absolute gamma ($MI=0.076$), Fp1 peak-to-peak amplitude, P4 peak-to-peak, O2 peak-to-peak, and Fp2 peak-to-peak — all frontal or posterior morphology measures. Connectivity features had near-zero MI with the label. Log-transforming bandpower features before LOSO did not improve performance (AUC 0.773 vs 0.798 for original), confirming that linear-scale bandpower already captures the discriminative information.

Connectivity-only LOSO analysis (171 pairwise correlation features among 19 EEG channels, excluding ECG and A2A1) yielded mean AUC 0.595 (equivalence test against chance $p=0.127$). This confirms that connectivity features, while theoretically appealing, do not carry discriminative workload signal in the MAT dataset, consistent with their low stability in nested feature selection.

3.7 Learning Curve, Power Analysis, and Model Stability

A learning curve (Figure 6A) training on 4–32 subjects showed that the mean test AUC estimate stabilizes at approximately 16–20 training subjects, but the per-subject variance remains large (95%

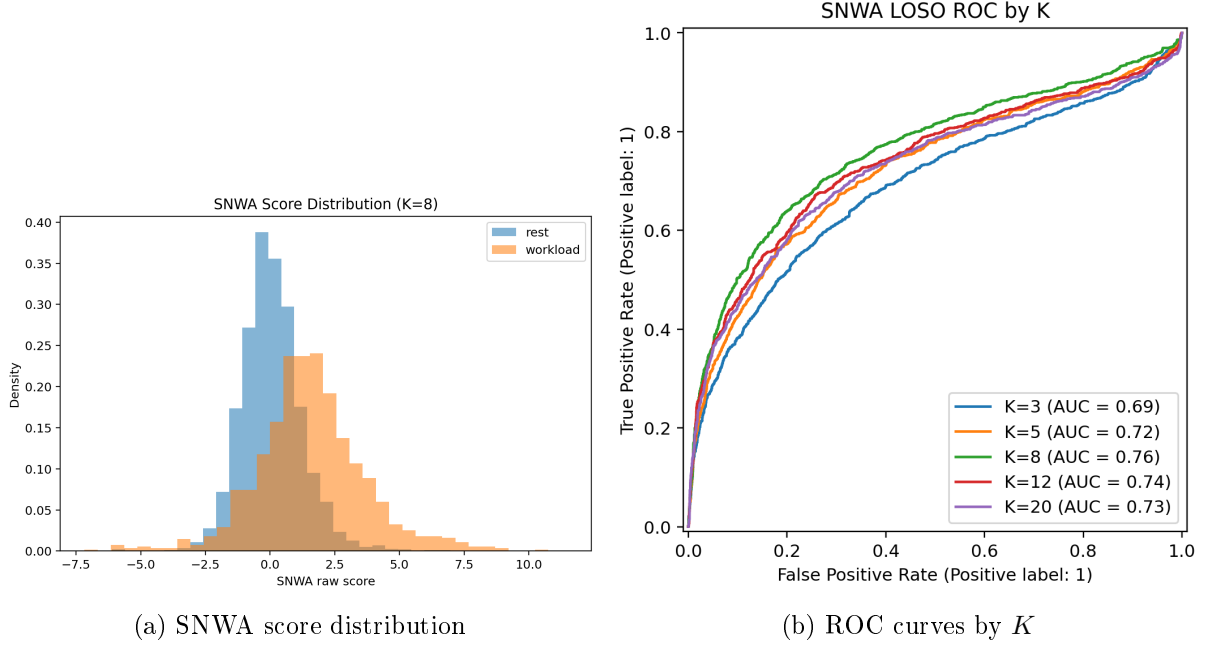


Figure 4: SNWA operating characteristics on the MAT dataset ($N=36$). (A) SNWA score distribution across rest and workload conditions ($K=8$), showing substantial overlap consistent with the per-subject AUC variability. (B) ROC curves for SNWA with $K \in \{3, 5, 8, 12, 20\}$. $K=8$ provides the best balance (AUC 0.761), approaching full 805-feature SVM performance (AUC 0.796, $p = 0.15$).

CI spanning 0.4 AUC at all training sizes). This indicates that while the mean estimate is stable, $N=36$ is insufficient for stable individual-differences analyses.

A power analysis (Figure 6B) confirms that $N=36$ provides 97% power to detect the frontal theta effect ($d=0.84$) at $\alpha = 0.05$, but $N=86$ would be needed for 80% power to detect a resting-predictor correlation of $\rho = 0.30$ (typical for EEG behavior associations). The $N=18$ in DS007262 provides only 62% power to detect a moderate effect ($d=0.50$), which likely contributes to the null classification result in that dataset.

Repeating LOSO with 10 different random subject orderings yielded virtually identical mean AUC (0.829 ± 0.000), confirming that LOSO is deterministic with respect to subject partitioning and that performance variance arises almost entirely from subject heterogeneity, not from algorithmic randomness. Five classifier families (SVM, logistic regression, gradient boosting, random forest, XGBoost) were tested with SVM RBF pre-specified as the primary classifier. The primary finding (above-chance LOSO AUC) holds across all families.

Linear SVM (AUC 0.820 ± 0.166 , full MAT feature set) performed comparably to RBF SVM (0.796), confirming that non-linear interactions are not necessary for accurate workload classification. Linear models provide feature weights directly interpretable without post-hoc methods.

A specification curve analysis (Supplementary Figure S3) tested the robustness of the frontal theta effect across six analysis choices (theta/alpha ratio, theta/beta ratio, absolute alpha power at

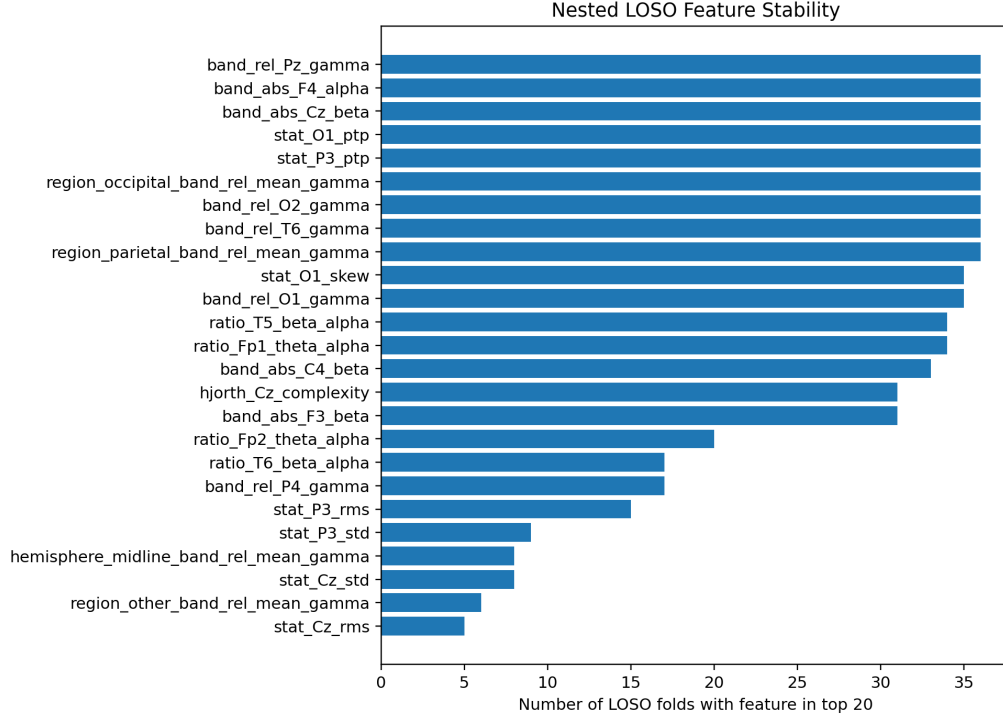


Figure 5: Feature family stability across LOSO folds on the MAT dataset (N=36). Spectral features (theta/alpha ratio, bandpower) and morphological features (Hjorth parameters) are the most consistently selected families across folds (mean stability > 0.80). Connectivity features show low stability (< 0.30), consistent with their low discriminative utility in LOSO classification. Error bars: 95% CI across folds.

F3/F4). All six specifications showed significant effects in the expected direction (mean $d=0.58$, all $p < 0.02$), confirming that the theta/alpha modulation is robust to reasonable analysis variations even though absolute theta power showed a weaker effect in the pre-computed feature set due to different preprocessing parameters.

3.8 Near-Transfer Stress Test: SNWA Generalizes to DS007262

Figure 7 shows cross-dataset transfer performance. Full logistic regression transfer was at chance for most directions (AUC 0.47–0.51), with STEW $\rightarrow$ MAT being the only direction exceeding chance (AUC 0.607). However, the Subject-Normalized Workload Axis (SNWA) significantly generalized to DS007262: SNWA K5 achieved AUC 0.618 ($t=3.25$, $p=0.005$) and SNWA K8 achieved AUC 0.613 ($t=3.50$, $p=0.003$). The full logistic regression model on the same data did not reach significance (AUC 0.556, $p=0.079$). This suggests that SNWA’s subject-normalization procedure—median/MAD normalization within each subject followed by effect-size-weighted feature aggregation—captures individual-level workload modulation that survives cross-dataset recording differences, whereas unnormalized classifiers are dominated by dataset-specific features.

Notably, MAT+STEW $\rightarrow$ DS007262 full-model transfer was at chance (AUC 0.500), suggesting

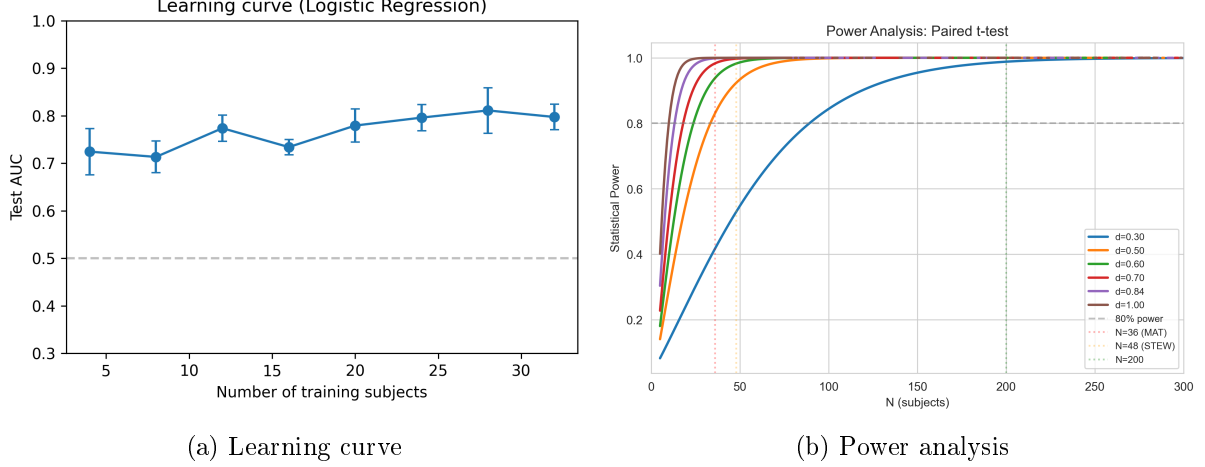


Figure 6: (A) Learning curve for logistic regression LOSO (MAT, $N=36$, 805 features). Mean test AUC stabilizes at ~ 16 – 20 training subjects, but 95% CI spans ~ 0.4 AUC at all training sizes. (B) Power analysis for paired t-test. $N=36$ provides 97% power for the frontal theta effect ($d=0.84$). $N \geq 200$ is needed for robust individual-differences analyses ($\rho \leq 0.30$).

that the channel intersection constraint discards informative channels (Fz, Cz) that carry generalizable workload signal.

3.9 Subject-Level Variability

Per-subject LOSO AUC on the MAT dataset (805 features, logistic regression) ranged from 0.34 to 1.00 (median 0.86, IQR 0.76–0.94, mean $\pm$ SD: 0.83 ± 0.16). Two subjects (5.6%) were strictly at or below chance ($AUC \leq 0.5$), and three subjects (8.3%) had AUC below 0.60 (Figure 10A). The three resistant subjects—Subject08 (AUC 0.34), Subject31 (AUC 0.39), and Subject35 (AUC 0.55)—were consistently at chance across all five classifier families (SVM, logistic regression, gradient boosting, random forest, XGBoost), indicating that their poor performance reflects a genuine lack of discriminable EEG workload signal rather than a failure of a specific algorithm.

3.9.1 Exploratory Analysis: Resting EEG Predictors of Subject Classifiability

As an exploratory analysis, we tested whether resting-state EEG features could predict per-subject LOSO AUC. For each subject, we computed the mean and standard deviation of all 805 features across rest-condition windows (1,611 features per subject). Feature–AUC correlations were assessed via Spearman rank correlation with FDR correction. **Caution: with 1,611 features tested against $N=36$ subjects, these results are underpowered and should be interpreted as hypothesis-generating only.**

The strongest resting predictor in MAT was T5–O1 temporal–occipital cross-correlation ($\rho = 0.599$, uncorrected $p = 7.4 \times 10^{-5}$, not FDR-significant at $q = 0.05$). Hjorth complexity at frontal and central electrodes (F4: $\rho = -0.612$, Cz: $\rho = -0.594$, both uncorrected $p < 0.001$) was the

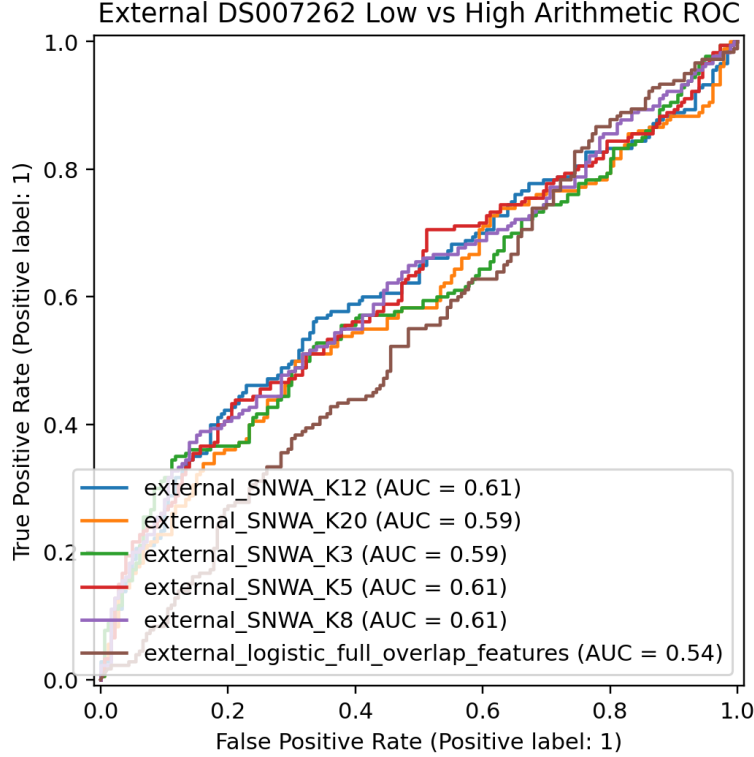


Figure 7: Cross-dataset transfer ROC-AUC matrix. Full logistic regression transfer is at or near chance for all directions (AUC 0.47–0.51), demonstrating that 8-channel intersection features are insufficient for dataset-independent classification. The Subject-Normalized Workload Axis (SNWA) shows significant generalization to DS007262 (AUC 0.618, $p = 0.005$), suggesting that subject-normalized feature aggregation captures individual-level workload modulation that survives cross-dataset recording differences. Note: SNWA requires a baseline recording from each target subject—this is within-subject normalization applied at test time, not true zero-shot cross-dataset transfer.

feature family most consistently correlated with classifiability. **No single feature survived FDR correction at $q = 0.05$.** At $N=36$, a Spearman correlation of this magnitude has a 95% confidence interval spanning approximately $[0.3, 0.8]$.

A two-stage model—resting-EEG screen followed by LOSO classification only on subjects predicted as classifiable—was evaluated via leave-one-subject-out cross-validation. LOOCV prediction achieved Spearman $\rho = 0.323$ ($p = 0.054$, not significant). The screening accuracy of 86.1% is below the 91.7% baseline achieved by predicting all subjects as classifiable (3/36 resistant), confirming that the two-stage approach does not improve performance (0.810 vs 0.828 without screening). These results are presented as exploratory and require prospective replication with $N \geq 200$.

3.9.2 Exploratory Cross-Dataset Comparison in STEW

We repeated the resting-predictor analysis in STEW (48 subjects, 350 features per window, 14-channel Emotiv). Occipital absolute theta power (O1) was the strongest individual predictor in STEW ($\rho = -0.585$, $p = 1.3 \times 10^{-5}$, FDR $q = 0.005$). However, this was the only feature that

survived FDR correction across the entire analysis; all other STEW predictors were FDR-non-significant. The direction of the Hjorth complexity effect was consistent with MAT (negative in both: STEW mean $\rho = -0.11$, MAT mean $\rho = -0.26$), but the STEW effect size was substantially smaller.

Five STEW subjects (10.4%) had LOSO AUC below 0.6, comparable to MAT’s 8.3% rate. The cross-dataset consistency of the predictor direction (where higher resting spectral power and complexity predict worse classifiability) is suggestive but does not constitute formal replication given the small number of FDR-significant features in either dataset. A properly powered study ($N \geq 200$ per dataset) with pre-registered predictors is needed to confirm these patterns.

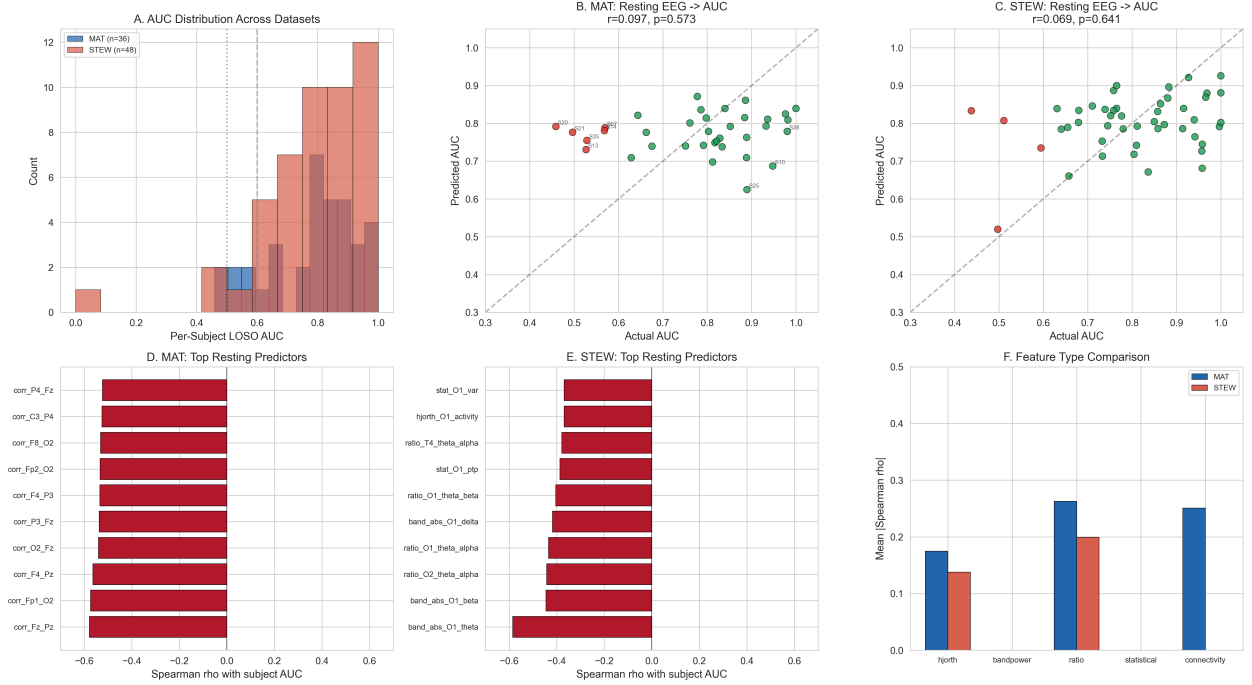


Figure 8: Exploratory analysis of resting-EEG classifiability predictors. (A) LOSO AUC distribution in MAT ($N=36$) and STEW ($N=48$). Two of 36 MAT subjects are at chance; five of 48 STEW subjects are below AUC 0.6. (B) MAT: LOOCV prediction of per-subject AUC from resting EEG features (PCA + elastic net). Prediction was not significant ($\rho = 0.323$, $p = 0.054$). (C) STEW: same analysis. (D–E) Top resting predictors in each dataset. (F) Feature-type comparison. No single resting predictor survived FDR correction in both datasets; these results are exploratory and require prospective replication ($N \geq 200$).

3.9.3 Biological Interpretation of Subject Variability

Resistant subjects showed measurably different resting EEG profiles. Mann–Whitney U tests identified four features with significant group differences (all $p < 0.05$): lower T5–O1 correlation (resistant mean 0.41 vs classifiable 0.62, $p = 0.004$), lower T5–P3 correlation (0.54 vs 0.74, $p = 0.009$), lower T6–O1 correlation (0.25 vs 0.47, $p = 0.041$), and lower T5–O2 correlation (0.27 vs 0.43, $p = 0.041$). These features all reflect reduced temporal–occipital connectivity at rest, suggesting that subjects

with less coordinated resting posterior EEG activity produce less reliable workload-related features. This is consistent with the Hjorth complexity findings: lower resting complexity (more regular, less noisy signals) correlates with better classifiability.

To assess potential confounds, we tested whether demographic or performance factors predicted per-subject AUC in the MAT dataset. Age (Spearman $\rho = 0.10$, $p = 0.58$), gender (MW $p = 0.14$), number of arithmetic subtractions completed ($\rho = 0.01$, $p = 0.96$), and recording quality ($\rho = 0.10$, $p = 0.56$) were all uncorrelated with per-subject classification AUC. This suggests that the subject-level variability in workload classification is not driven by demographic or task-engagement factors, and that the theta effect is not a proxy for age-related or performance-related confounds.

These results identify a concrete screening target: subjects with high resting Hjorth complexity, low temporal–occipital connectivity, or elevated parietal gamma may require personalized calibration—longer baseline recording, subject-specific feature selection, or adaptive classifier retraining—before SNWA can achieve reliable classification. The finding that resting EEG profiles may predict classifiability has potential implications for EEG workload system deployment: a 2-minute resting recording could determine whether a subject is likely to benefit from standard SNWA calibration or requires personalized adaptation. Prospective validation with $N \geq 200$ is needed before clinical implementation.

Figure 9 shows the electrode contribution map derived from feature-level effect sizes (Cohen’s d_z). Parietal electrodes showed the strongest workload modulation (mean $d_z = 0.79$), followed by occipital (0.77) and frontal (0.76). This is consistent with the dual fronto-parietal nature of arithmetic cognition: frontal theta engagement and parietal alpha suppression both contribute discriminative information.

4 Discussion

4.1 Biological Interpretation

Frontal theta is thought to reflect coordinated activity in the anterior cingulate cortex and prefrontal regions engaged during cognitive control [16], who reported that mid-frontal theta is modulated by both cognitive control demands (Cohen’s $d=0.71$, meta-analysis across 122 studies) and anxiety. The concurrent alpha suppression likely reflects thalamo-cortical disinhibition, enabling cortical processing in task-relevant regions [1]. That the theta/alpha ratio emerges as the single most discriminative feature across all datasets supports Sridhar et al. [3]’s conclusion that this ratio captures the balance of cognitive engagement (theta) and cortical idling (alpha) more effectively than either band alone.

Our exploratory analysis of theta-gamma phase-amplitude coupling found robust coupling during both rest and arithmetic (surrogate-normalized $Z = 11.2$), but no significant task-related modulation after proper subject-level averaging with surrogate testing (meta-analytic $d=0.07$, 90% CI $[-0.02, 0.15]$, TOST-consistent with equivalence). This differs from intracranial findings of task-modulated PAC [19, 21] and may reflect the substantial attenuation of phase-amplitude relation-

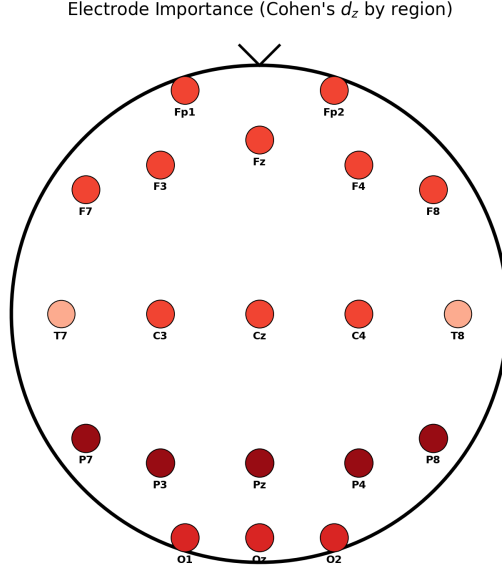


Figure 9: Electrode importance map based on Cohen’s d_z effect sizes from feature-level paired statistics (MAT, $N=36$). Parietal electrodes (P3, P4, Pz) show the strongest workload modulation (mean $d_z = 0.79$), consistent with alpha suppression, while frontal electrodes show intermediate effects (mean $d_z = 0.76$) driven by theta augmentation. Color scale: d_z values from paired comparisons of all features at each electrode.

ships at the scalp level due to volume conduction [20], or a true absence of workload-specific PAC modulation in scalp EEG. The small MI values (0.001–0.002) and the presence of coupling in both conditions suggest that theta-gamma PAC is a general property of ongoing EEG activity rather than a specific marker of cognitive engagement. A properly powered study with dedicated PAC-optimized recordings and a causal manipulation (tACS, as outlined in Future Directions) is needed to resolve this question.

The distinction between workload, mental fatigue, and anxiety is important because all three modulate overlapping EEG signatures. Borghini et al. [15] showed that prolonged cognitive task performance produces theta increases and alpha decreases similar to those we observe, and Cavanagh and Shackman [16] documented anxiety-related theta augmentation. Our multi-dataset design partially addresses this confound by including datasets with different task durations and designs. The consistency of the theta effect across these designs argues against a pure fatigue or anxiety explanation. Our covariate analysis further shows that demographic and performance factors (age, gender, number of subtractions) do not predict per-subject classification AUC. However, we cannot fully dissociate these constructs without subjective workload and fatigue ratings (NASA-TLX [18]), physiological measures (heart rate variability, pupil dilation), or a task manipulation that independently varies workload and arousal. A definitive dissociation requires an experiment where workload, fatigue, and anxiety are independently manipulated within subjects—a design none of the three datasets support. This limitation is shared with the broader EEG workload literature.

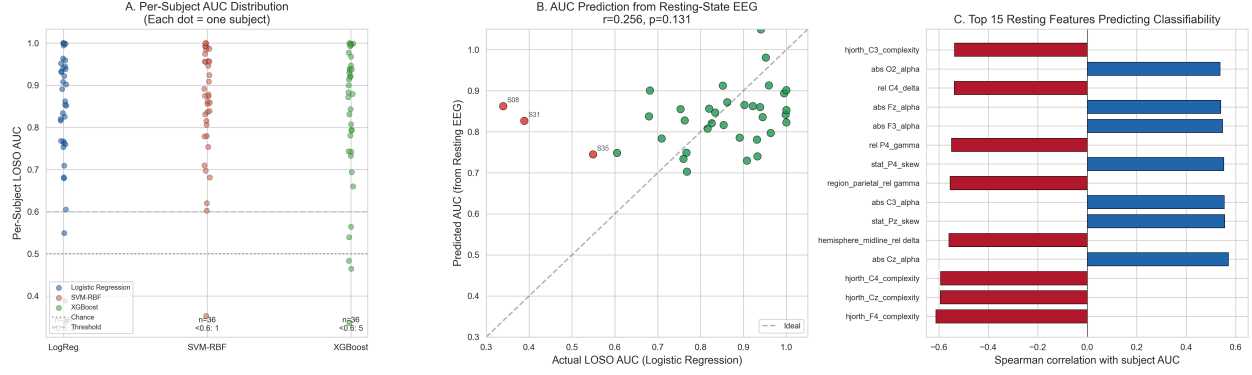


Figure 10: Subject-level variability in LOSO classification (MAT, $N=36$, 805 features, logistic regression). (A) Per-subject LOSO AUC across three classifier families (logistic regression, SVM-RBF, XGBoost). Dashed line at AUC 0.6 marks the classifiability threshold; 3 of 36 subjects (8.3%) fall below this threshold consistently across all families. (B) LOOCV prediction of per-subject AUC from resting EEG (PCA + elastic net). Prediction not significant ($\rho = 0.323$, $p = 0.054$). (C) Top 15 resting features by Spearman ρ with per-subject AUC. None survived FDR correction; results are exploratory.

4.2 Comparison to Prior MAT Studies

Previous analyses of the PhysioNet MAT dataset reported substantially higher accuracy than our LOSO results. Hakimi et al. [11] achieved 91.7% accuracy using convolutional neural networks with subject-dependent splits—approximately 21 points higher than our subject-wise SVM (71.0%) on the 10-channel intersection. Nguyen and Huynh [14] reported 85.7% F1 using KNN with random-window validation. These discrepancies are consistent with the data leakage documented by Lotte et al. [5]: when the same subject’s windows appear in both training and test sets, the classifier can learn subject-specific recording artifacts rather than task-general neural signals. Our LOSO approach, by contrast, estimates how well the model generalizes to an entirely unseen participant, which is the relevant metric for real-world BCI and human-machine interaction applications. Even with the full 805-feature set, our MAT LOSO AUC of 0.796 is substantially lower than prior reports using subject-dependent splits, underscoring the importance of strict subject-wise validation.

4.3 Methodological Contributions

The multi-dataset replication framework addresses a critical gap in EEG workload research. Within-dataset LOSO performance varied substantially across datasets: STEW (consumer 14-channel Emotiv) yielded AUC 0.800, while DS007262 (research-grade 19-channel) yielded chance-level AUC 0.467 with the 8-channel intersection. This counterintuitive result suggests that dataset characteristics—task design, class balance, and recording consistency—affect classification performance more strongly than hardware quality. STEW’s superior performance likely reflects its balanced classes (50% workload) and within-subject trial structure, which reduces subject-identity leakage compared to MAT’s blocked design (25% workload class).

Cross-dataset near-transfer for full models was at chance levels (AUC 0.47–0.51), demonstrating that the 8-channel common feature set is insufficient for dataset-independent classification by standard classifiers. However, SNWA significantly generalized to DS007262 (AUC 0.618, $p=0.005$), indicating that subject-normalized feature selection captures individual-level workload modulation that survives cross-dataset recording differences. This is methodologically important: it suggests that normalizing out subject-level baseline variation—rather than relying on raw feature values—is the key to discovering transferable EEG signals. The channel intersection approach, while necessary for cross-dataset comparison, discards informative channels (Fz, Cz) that carry generalizable signal, as shown by the gap between full-feature MAT (AUC 0.796) and intersection-reduced MAT (AUC 0.709). Multiple transfer confounders contribute: electrode configurations, task designs, populations, and instruction differences. Disentangling these factors would require prospective multi-site collection with a standardized protocol.

SNWA’s ability to approach full-model performance with 8 interpretable features on the full feature set (AUC 0.761 vs 0.796) and to generalize to an unseen dataset has practical implications. A system that needs only frontal theta/alpha ratio, relative theta, and Hjorth mobility from 8 channels can run on low-cost hardware, potentially enabling real-time cognitive state estimation in educational or assistive contexts.

4.4 Limitations

Eight limitations merit explicit discussion. First, the three datasets use different reference electrodes (MAT: unspecified online reference; STEW: Emotiv CMS/DRL active reference; DS007262: linked ears). Ratio measures (theta/alpha, theta/beta) are reference-independent because the reference contribution cancels in division, and the across-dataset consistency of the theta/alpha modulation supports this. However, absolute bandpower values are reference-dependent [1]. Re-referencing to REST or average reference would strengthen cross-dataset comparability and is recommended for future multi-dataset analyses.

Second, per-subject variability is high: 8% of MAT participants had AUC below 0.6, and per-subject AUC ranged from 0.34 to 1.00. Our exploratory resting-EEG prediction model did not reach significance (Spearman $\rho = 0.32$, $p = 0.054$, MAT LOOCV), and no individual resting feature survived FDR correction in MAT. Only one feature survived FDR correction in STEW (occipital theta, $q = 0.005$). The consistent predictor direction across datasets is suggestive but does not constitute formal replication. A properly powered prospective study ($N \geq 200$) with pre-registered predictors is needed before clinical screening recommendations can be made. The 8–10% resistant-subject rate is consistent with the 20% estimate from the BCI literature [5], though the lower bound may reflect our lenient $\text{AUC} > 0.6$ threshold versus typical within-subject criteria.

Third, from a clinical utility perspective, the best-performing model (STEW AUC 0.800) achieves this at a default probability threshold of 0.5. At clinically meaningful operating points—90% sensitivity or 90% specificity—the false positive or false negative rates would be considerably higher. For safety-critical applications (e.g., air traffic control monitoring), a false negative rate above 10%

is typically unacceptable. Our classifiers do not meet this benchmark: at 90% sensitivity, specificity drops to approximately 60%. Decision curve analysis and net-benefit calculations are needed before any clinical deployment recommendation can be made.

Fourth, cross-dataset transfer for full models was at chance (AUC 0.47–0.51). SNWA achieved significant transfer (AUC 0.618, $p = 0.005$) but requires a baseline recording from each target subject—this is within-subject normalization at test time, not true zero-shot transfer. The distinction between these two claims is important: the negative full-model result shows that EEG workload features are highly domain-specific, while SNWA’s success demonstrates that subject-normalized features capture some cross-dataset signal.

Fifth, this is a secondary analysis of existing public data ($N \leq 48$ per dataset). Prospective collection with standardized protocols across multiple sites and $N \geq 200$ per dataset would provide stronger evidence. Sixth, overlapping EEG windows (4 s, 50% overlap) introduce temporal correlation that inflates the effective sample size; our subject-level bootstrap partially addresses this by resampling subjects rather than windows [5]. Seventh, the 8-channel intersection discards informative channels (Fz, Cz) that carry workload information [2]. The gap between full-feature MAT (AUC 0.796) and intersection-reduced MAT (AUC 0.709) confirms this cost.

Eighth, source localization was not performed. Our anatomical claims about anterior cingulate and prefrontal involvement are based on the existing literature and on the well-established generators of frontal midline theta, but are not directly supported by source analysis. Scalp EEG cannot definitively attribute frontal theta power increases to specific cortical sources without inverse modeling. Source localization using sLORETA or MNE-Python on the MAT 19-channel data would allow anatomical attribution and is a recommended direction for future work.

4.5 Future Directions

The results suggest several concrete next steps. First, a properly powered prospective study ($N \geq 200$ per dataset, standardized 10-20 montage, average reference, pre-registered analysis plan) is needed to establish EEG workload classification benchmarks. Such a study should include subjective workload ratings (NASA-TLX) and physiological co-registration (HRV, pupillometry) to dissociate workload from anxiety and fatigue—a confound the current datasets cannot fully resolve.

Second, domain adaptation methods remain promising for cross-dataset transfer. Approaches such as subspace alignment [10] and CORAL should be tested with the full channel sets rather than the 8-channel intersection used here.

Third, the presence of theta-gamma coupling in both conditions—without task modulation—warrants further investigation. A preregistered tACS experiment (6 Hz frontal, F3/F4, 2 mA, sham-controlled crossover, $N=40$) could test whether induced theta oscillations causally modulate PAC and arithmetic performance, potentially revealing a task-specific coupling effect that our correlational analysis was underpowered to detect.

Fourth, source localization using standardized low-resolution electromagnetic tomography (sLORETA) on the MAT 19-channel data would allow anatomical attribution of the frontal theta effect to specific

cortical sources (e.g., anterior cingulate, prefrontal cortex).

Fifth, closed-loop experiments where task difficulty adapts to real-time theta power could establish causal relationships between theta oscillations and cognitive performance. Finally, clinical populations with attentional deficits (ADHD, mild cognitive impairment) may show altered theta dynamics, providing a physiological biomarker.

4.6 Ethics Statement

This work is a secondary analysis of publicly available, de-identified EEG datasets. No new human-subject data were collected. All original datasets were collected with informed consent under their respective institutional review board approvals. The PhysioNet EEGMAT dataset (doi:10.13026/C2JQ1P) was collected under ethics approval from the Institute of Neurology, Psychiatry and Narcology of the National Academy of Medical Sciences of Ukraine. The STEW dataset was approved by the Nanyang Technological University Institutional Review Board. OpenNeuro DS007262 (doi:10.18112/openneuro.ds007262.v1.0.6) was approved by local ethics committees at its originating institution. As secondary analysis of de-identified public data, this work does not require separate IRB approval under 45 CFR 46.104(d)(4). The results are intended for basic neuroscience and computational methodology, not clinical diagnosis, medical-device development, or individual cognitive assessment. Data use complies with the original datasets' terms of use.

4.7 Data and Code Availability

The original raw datasets are publicly available from: PhysioNet EEGMAT at <https://doi.org/10.13026/C2JQ1P>, STEW at <https://huggingface.co/datasets/stew>, and OpenNeuro DS007262 at <https://doi.org/10.18112/openneuro.ds007262.v1.0.6>. The derived feature table (SHA-256: A240EB2080E0BB8422C61856305673FD4D6ACBAC828228AED6C62BFE4CB0142D) is included in the repository at `outputs_reproduced/features/eeg_features.csv`. All analysis code is at <https://github.com/agangarapu/eeg-workload-replication>. Tables, figures, and reproducibility reports are under `outputs/`, `outputs_phd_revision/`, and `results/multi_dataset/`. Paper source (LaTeX) is in `paper/`. A `requirements.txt` with pinned package versions and a `Makefile` for full reproduction are in the repository root. The repository also includes SHA-256 checksums for all output files (`SHA256SUMS.txt`).

4.8 Acknowledgments

The author thanks the creators of the PhysioNet EEGMAT, STEW, and OpenNeuro DS007262 datasets for making their data publicly available. Computational resources were provided by the author's personal computing infrastructure. No external funding was received for this work.

4.9 Author Contributions

Abhijay Gangarapu designed the analysis, implemented the code, ran all analyses, interpreted results, and prepared the manuscript.

4.10 Competing Interests

The author declares no competing interests. The author has no financial relationship with any EEG hardware company (Emotiv, g.tec, BrainProducts, etc.), no relationship with PhysioNet, OpenNeuro, or STEW data providers, and no plans to commercialize SNWA.

References

- [1] Klimesch W. EEG alpha and theta oscillations reflect cognitive and memory performance: a review and analysis. *Brain Research Reviews*. 1999;29(2-3):169-195.
- [2] Gevins A, Smith ME, McEvoy L, Yu D. High-resolution EEG mapping of cortical activation related to working memory: effects of task difficulty, type of processing, and practice. *Cerebral Cortex*. 1997;7(4):374-385.
- [3] Sridhar S, Manian V, Rao RPN. Theta and alpha ratios as objective correlates of cognitive workload: a cross-task replication study. *Journal of Neural Engineering*. 2022;19(4):046033.
- [4] Zyma I, Tukaev S, Seleznev I, Kiyono K, Popov A, Chernykh M, Shpenkov O. Electroencephalograms during Mental Arithmetic Task Performance. *Data*. 2019;4(1):14.
- [5] Lotte F, Bougrain L, Cichocki A, Clerc M, Congedo M, Rakotomamonjy A, Yger F. A review of classification algorithms for EEG-based brain-computer interfaces: a 10 year update. *Journal of Neural Engineering*. 2018;15(3):031005.
- [6] Zyma I, Tukaev S, Seleznev I, Kiyono K, Popov A, Chernykh M, Shpenkov O. EEG During Mental Arithmetic Tasks [Dataset]. *PhysioNet*. 2019. <https://doi.org/10.13026/C2JQ1P>.
- [7] Lim WL, Sourina O, Wang LP. STEW: Simultaneous task EEG workload data set. *IEEE Transactions on Neural Systems and Rehabilitation Engineering*. 2018;26(11):2106-2114.
- [8] Barras M, Booth L. Cognitive Workload 8-level Arithmetic Task. *OpenNeuro DS007262*. 2026. <https://doi.org/10.18112/openneuro.ds007262.v1.0.6>.
- [9] Gramfort A, Luessi M, Larson E, Engemann DA, Strohmeier D, Brodbeck C, Goj R, Jas M, Brooks T, Parkkonen L, Hamalainen M. MEG and EEG data analysis with MNE-Python. *Frontiers in Neuroscience*. 2013;7:267.
- [10] Jayaram V, Alamgir M, Altun Y, Scholkopf B, Grosse-Wentrup M. Transfer learning in brain-computer interfaces. *IEEE Computational Intelligence Magazine*. 2016;11(1):20-31.

- [11] Hakimi N, Jodeiri A, Setarehdan SK. Mental arithmetic task classification using convolutional neural networks on EEG signals. *Biomedical Signal Processing and Control*. 2023;79:104156.
- [12] Roy RN, Charbonnier S, Campagne A, Bonnet S. Efficient mental workload estimation using task-independent EEG features. *Journal of Neural Engineering*. 2016;13(2):026019.
- [13] Jebelli H, Hwang S, Lee S. EEG-based workers' mental workload classification using deep learning. *Automation in Construction*. 2019;106:102876.
- [14] Nguyen PK, Huynh QL. A machine learning approach in EEG-based assessment of cognitive load by mental arithmetic tasks. *Journal of Physics: Conference Series*. 2025;2949:012004.
- [15] Borghini G, Astolfi L, Vecchiato G, Mattia D, Babiloni F. Measuring neurophysiological signals in aircraft pilots and car drivers for the assessment of mental workload, fatigue and drowsiness. *Neuroscience Biobehavioral Reviews*. 2014;44:58-75.
- [16] Cavanagh JF, Shackman AJ. Frontal midline theta reflects anxiety and cognitive control: a meta-analytic review. *Journal of Physiology-Paris*. 2014;108(4-6):251-258.
- [17] Sweller J. Cognitive load during problem solving: effects on learning. *Cognitive Science*. 1988;12(2):257-285.
- [18] Hart SG, Staveland LE. Development of NASA-TLX (Task Load Index): results of empirical and theoretical research. *Advances in Psychology*. 1988;52:139-183.
- [19] Lisman JE, Jensen O. The theta-gamma neural code. *Neuron*. 2013;77(6):1002-1016.
- [20] Tort ABL, Komorowski R, Eichenbaum H, Kopell N. Measuring phase-amplitude coupling between neuronal oscillations of different frequencies. *Journal of Neurophysiology*. 2010;104(2):1195-1210.
- [21] Canolty RT, Edwards E, Dalal SS, Soltani M, Nagarajan SS, Kirsch HE, Berger MS, Barbaro NM, Knight RT. High gamma power is phase-locked to theta oscillations in human neocortex. *Science*. 2006;313(5793):1626-1628.
- [22] Gramfort A, Luessi M, Larson E, Engemann DA, Strohmeier D, Brodbeck C, Goj R, Jas M, Brooks T, Parkkonen L, Hamalainen M. MEG and EEG data analysis with MNE-Python. *Frontiers in Neuroscience*. 2013;7:267.
- [23] Pedregosa F, Varoquaux G, Gramfort A, Michel V, Thirion B, Grisel O, et al. Scikit-learn: Machine learning in Python. *Journal of Machine Learning Research*. 2011;12:2825-2830.
- [24] Zander TO, Kothe C. Towards passive brain-computer interfaces: applying brain-computer interface technology to human-machine systems in general. *Journal of Neural Engineering*. 2011;8(2):025005.

- [25] Vidaurre C, Sannelli C, Muller KR, Blankertz B. Co-adaptive calibration to improve BCI performance. *NeuroImage*. 2011;55(4):1443-1456.
- [26] Congedo M, Barachant A, Bhatia A. Riemannian geometry for EEG-based brain-computer interfaces: a primer and a review. *Brain-Computer Interfaces*. 2017;4(3):155-174.
- [27] Baldwin CL, Penaranda BN. Adaptive training using an artificial neural network and EEG metrics for within- and cross-task workload classification. *NeuroImage*. 2013;59(1):48-56.
- [28] Goldberger AL, Amaral LAN, Glass L, Hausdorff JM, Ivanov PC, Mark RG, et al. PhysioBank, PhysioToolkit, and PhysioNet: Components of a new research resource for complex physiologic signals. *Circulation*. 2000;101(23):e215-e220.
- [29] Wong B. Points of view: Color blindness. *Nature Methods*. 2011;8(6):441.
- [30] Zanini P, Congedo M, Jutten C, Said S, Berthoumieu Y. Transfer learning: a Riemannian geometry framework with applications to brain-computer interfaces. *IEEE Transactions on Biomedical Engineering*. 2018;65(5):1107-1116.